Probability distribution

Distribution
↳ The possible values a variable can take and how frequently they occur

$P(Y) \rightarrow$ ~~all~~ possible outcome of event or actual outcome

$P(y) \rightarrow$ one of the possible outcomes

$P(Y=y)$

probability measure the likelihood of an outcome

$\sigma^2 = E(Y^2) - \mu^2$

Types of distribution:-

1] discrete distribution
↳ has a finite number of outcomes

2] continuous distribution
↳ infinitely many outcomes

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types of discrete distribution :-

1 uniform distribution ($U(a,b)$)

$X \sim U(3,7)$ كل الاحتمالات متشابهة
→ each out come is equally likely to happen
→ Both mean and variance are uninterpretable and gives no predictive power

2 Bionomial distribution
Notation
~~Bernollie~~ : $X \sim B(n, p)$ → ~~Bernollie~~
number of triles
احتمالية نجاح كل تجربة فردية

~~برنوللي~~ Binomial بيكون more than one trial

الكل ← ~~Binomial~~ Bernollie ← زي ~~برنوللي~~ Binomial بس لتجربة وحدة

Binomial = $B(1, p)$
↳ a sequence of identical Bernoulli events

$Bern(B) = B(1, p)$

احتمال السؤال الأول في الامتحان يكون صح ← برنوللي
احتمال الاسئلة كلها صح ← Binomial

E(Bernoulli event) → Which outcome we expect for a single trial

E(Binomial event) The number of times we expect to get a specific outcome

* Binomial probability function:

$P(y)$ prop

$p(\text{desired outcome}) = P$

$p(\text{alternative outcome}) = 1-P$

$$P(y) = \binom{n}{y} \cdot P^{y} \cdot (1-P)^{n-y}$$

$E(x) = n \cdot P$ expected Val.

$\sigma^2 = n \cdot P(1-P) = E(y^2) - E(y)^2$ variance

* لما يقولي في مسألة احتمالية لحدث معين بحسب الاحتمالية له بس
* at least (r) ← بعد وبحسب التوافيق من r لحد n و اجمع
* at most (r) ← بعد وبحسب من r لحد صفر

* Bernoulli distribution :-

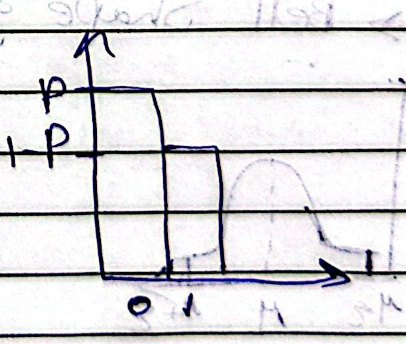
Bern (P) , $x \sim \text{Bern}(P)$

used For events with:

1. 1 Trial
2. with 2 possible outcomes

$E(x) = P$ or $1-P$

$\sigma^2 = P(1-P)$



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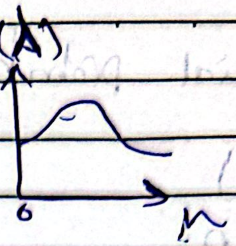
No: Continuous dist.

* Poisson distribution



(ex)

$Po(\lambda)$ $\quad Y \sim Po(y)$

$\lambda \rightarrow$ average value most of the time it's known

$$P(y) = \frac{\lambda^{y} e^{-\lambda}}{y!}$$

$e \rightarrow$ constant $= 2.7$

$E(y) = \lambda$

$\sigma^2 = \lambda$

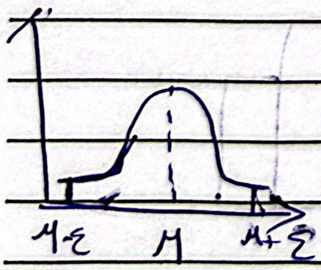
* Normal distribution

$N(\mu, \sigma^2)$

→ Follows natur (ex): the size of a full-grown male Lion 150 - 250 kg

لو قلهم 150 > أو أكبر 250 < بس قيم نادرة or (outliers)

→ Bell Shape graph (seymmetric)



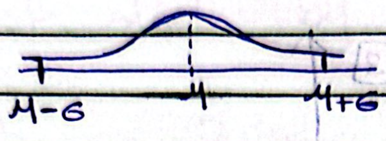
* $E(x) = \mu$

$Var(x) = E(x^2) - E(x)^2$

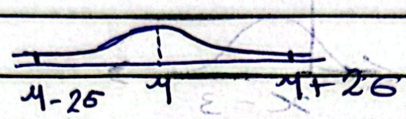
* 68, 95, 99.7 Law

* for all normally distributed events, 68% of all out comes falls within 1 standard deviation away from the mean, 95% fall within 2 standard deviations and 99.7 within 3

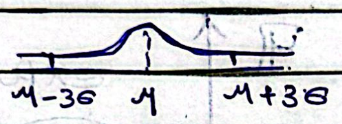
68% →



95% →



99.7 →



* outliers are extremely rare.

standard normal distribution

Transformation:
↳ A way in which we can alter every element of a distribution to get a new distribution

→ works on large ~~..~~ sized data

لو عندي ~~....~~ نورمال dist $x \sim N(\mu, \sigma^2)$

طب $x+3 \sim N(\mu, \sigma^2)$ أو $x^2 \sim N(\mu, \sigma^2)$ ،

دي أحسبها ازاي دة ← بقول للمنحنى الجرسي تحويل

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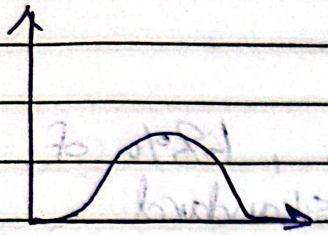


$x \sim N(\mu, \sigma^2)$

or $y = f(x)$

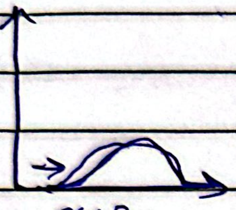
1] $y = f(x+3)$

2] $y = f(x-3)$

3] $y = f(2x)$

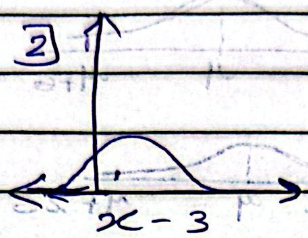
4] $y = f(\frac{x}{2})$

1]



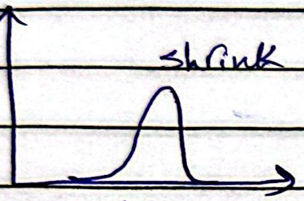
$x+3$

2]



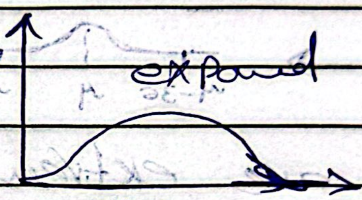
$x-3$

3]



$f(x \cdot c)$

4]



$f(\frac{x}{c})$

Note: لو قسمت على كسر هـ يقلب ضرب

* standardizing

→ special case of transformation in which we make the expected value $E(x)$ equal to zero, and the variance equal to one

* standardizing normal distribution is called standard Normal distribution

بخلينا نحول أي قيمة من التوزيع الطبيعي لـ z-score

where $z = \frac{x - \mu}{\sigma}$

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* steps for standard normal dist.

1] First we move the graph either to the left or Right till it's mean equals zero by $(x-\mu)$

2] make sure that STD = 1 = σ by dividing every element of dist by the STD = σ

3] $Z \sim N(0,1)$
$y \sim N(\mu, \sigma^2)$ $\Big\} \quad z = \frac{y-\mu}{\sigma} \longrightarrow z$

* applying this trans. for Normal dist. would result in a standard normal dist.

* every element of the non-Standardized dist. Is represented in the new distribution by the number of standard deviations away from the mean

Student's T dist.

⇒ a small sample size approximation of a Normal Distribution

$t(k)$ ← degrees of freedom

ولانها بتشتغل على sample بتكون نسبه الخطأ كبيرة ف t بيعالج المشكلة لانها:

← أعرض من التوزيع الطبيعي

← الاحتمالات فيه اكبر عند الأطراف "tails" علشان يعوض عدم الدقة في العينة الصغيرة.

to accommodate the occurrence of values ~~fare~~ away from the mean

if $k > 2$

$$\Rightarrow \quad E(y) = \mu$$

$$Var(y) = \frac{(\underbrace{s^2}_{\text{var of sample}} \cdot k)}{k-2}$$

→ used for

1) frequently used when conducting statistical analysis

2) Hypothesis testing with limited data

3) CDF table (T-table)

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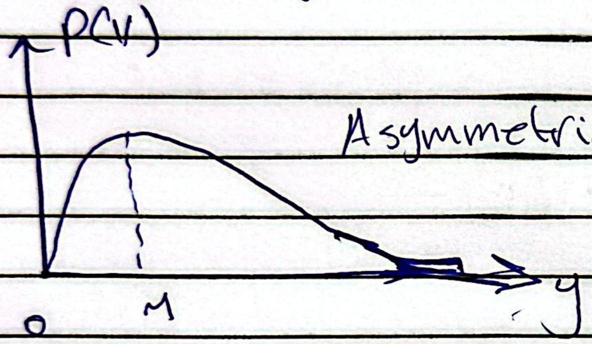
Chi ~~distribution~~ Squared distribution.

$\chi^2(K)$ — Freedom degree (K)
$Y \sim \chi^2(K)$

* very few events in real life follow Chi squared dist.

* most used for statistical analysis in → Hypothesis testing
→ computing confidence intervals

also found when determining the goodnes of fit of categorical values.



$Y \sim t(K)$
$Y^2 \sim \chi^2(K)$

, $x \sim \chi^2(K)$
$\sqrt{x} \sim t(K)$

⇒ contains a table of known values: N, T

$E(X) = K$, $Var(X) = 2K$

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Conditional Probability

$$P(A|B) = \frac{P(AB)}{P(B)}$$

$P(AB)$ ← comanded

$P(B)$ ← the condition

* multiplication Rule :

E_1 = short E_2 = smart

$P(\cancel{\ } \text{short}, \text{smart}) = P(\text{short})\, P(\text{smart}|\text{short})$

$$P(ABC) = P(A)\, P(B|A)\, P(C|AB)$$

* Bayes's Formula :

$$P(A|B) = \frac{P(B|A)\, P(A)}{P(B)}$$

العكس

وأحط الثاني جيه والأول تحته

* odds → How likely somthing to happen VS. not happen

$$\frac{P(A)}{P(A^c)}$$

$P(A)$ → happen

$P(A^c)$ → don't happen

(ex) what are the odds you Draw pink or orange?

5pink 3orange 7green

$P(A) = \frac{8}{15}$

$P(A^c) = \frac{7}{15}$

$$\frac{P(A)}{P(A^c)} = \frac{8}{7}$$

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* Independence :-

$\rightarrow P(AB) = P(A)\,P(B)$

or

$\Rightarrow P(A|B) = P(A)$ and $P(B|A) = P(B)$

$P(ABCD\ldots) = P(A)P(B)P(C)P(D)$

(ex)

1] If you flip 2 coins Is the first flip Being Heads Independent to the second flip Being Heads? يعني هل ما في تأثير للأولى عالثانية

yes

A = first flip Heads
B = second flip Heds

$S = \{\boxed{(H,H)}, (H,T), (T,H), (T,T)\}$

$P(AB) = 1/4$

2] If The first flip Being Heads independent to the Number of Heads Being odd?

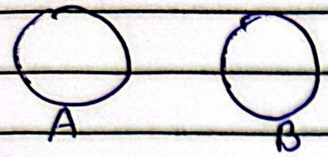
بدي نجربه واكتب الاحتمالات وأضربها في بعض

A = first flip heads
B = heads odd

$\Rightarrow P(AB) = P(A)P(B)$

$\frac{1}{4} = \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{4}$ ✓

* mutually exclusive evens



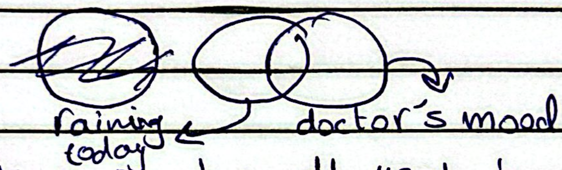
ـ يعني مفيش حاجة مشتركة بينهم وغير متقاطعين

→ two events can't occur together

$P(A \cap B) = 0$

* independent events

→ the occurence of one event doesn't affect the occurence of another event



ملهمش علاقة ببعض بس في احتمال الجو يمطر واحنا بناخد محاضرة والدكتور يكون متعصب فيها

$$P(A \cap B) = P(A) \cdot P(B)$$

$P(A|B) = P(A) \rightarrow$ independent A & B